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Scalable Scanning and Automatic Classification of TLS Padding Oracle Vulnerabilities

Robert Merget and Juraj Somorovsky, *Ruhr University Bochum*; Nimrod Aviram, *Tel Aviv University*; Craig Young, *Tripwire VERT*; Janis Fliegenschmidt and Jörg Schwenk, *Ruhr University Bochum*; Yuval Shavitt, *Tel Aviv University*

The TLS protocol provides encryption, data integrity, and authentication on the modern Internet. Despite the protocol's importance, currently-deployed TLS versions use obsolete cryptographic algorithms which have been broken using various attacks. One prominent class of such attacks is CBC padding oracle attacks. These attacks allow an adversary to decrypt TLS traffic by observing different server behaviors which depend on the validity of CBC padding.

We present the first large-scale scan for CBC padding oracle vulnerabilities in TLS implementations on the modern Internet. Our scan revealed vulnerabilities in 1.83% of the Alexa Top Million websites, detecting nearly 100 different vulnerabilities. Our scanner observes subtle differences in server behavior, such as responding with different TLS alerts, or with different TCP header flags.

We used a novel scanning methodology consisting of three steps. First, we created a large set of probes that detect vulnerabilities at a considerable scanning cost. We then reduced the number of probes using a preliminary scan, such that a smaller set of probes has the same detection rate but is small enough to be used in large-scale scans. Finally, we used the reduced set to scan at scale, and clustered our findings with a novel approach using graph drawing algorithms.

Contrary to common wisdom, exploiting CBC padding oracles does not necessarily require performing precise timing measurements. We detected vulnerabilities that can be exploited simply by observing the content of different server responses. These vulnerabilities pose a significantly larger threat in practice than previously assumed.

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